

up SMT lines, importing products in the CKD status for assembling and selling them in India. There are also big and small companies that even export these out of India after assembling them here and giving them the 'Made in India' tag.

"This is where start-ups can come in, do innovation and become the game changers for the country. The talent that is working in global captive semiconductor design centres can come in and help create a fabless ecosystem for semiconductors in the country. That is how 40 percent of the IP and technology value can be realised within the country and we can do a much better value addition than what we are doing already," explains Keskar.

There is a strong chance that most of the professionals and institutions will read everything mentioned by Keskar as a big challenge. But these challenges are the ones that are the right set of opportunities up for grabs for MSMEs and electronics start-ups in the country.

The government of India, as mentioned earlier, has already announced many policies focused on electronics manufacturing in the country. While the component manufacturing ecosystem will take around five years, the low hanging fruit—creating IP and technology—can be started to be worked upon from this very moment.

"Creating products and innovation is the need of the hour and start-ups can play a huge role here," adds Keskar.

Electronics, in terms of import bills, capture the second position after oil. India's electronics imports, in 2019 were worth \$55 billion, and the exports were around \$8.8 billion. According to the Commerce ministry, the exports of consumer electronics goods almost doubled to \$11.8 billion in 2020 from \$6.4 billion in 2018.

The challenges and opportunities

Though there are a lot of opportunities up for grabs for the MSMEs and electronics start-ups, it still is not going to be a walk in the park for them. The government of India, itself, did not shy away from admitting that the IT hardware manufacturing capability

and capacity in the country has progressively declined, and many units have either ceased operations or are operating at low capacities.

One of the PLI schemes reads, "The IT Hardware manufacturing sector faces the lack of a level playing field vis-à-vis competing nations. As per industry estimates, the electronics manufacturing sector suffers from a disability of around 8.5 percent to 11 percent on account of lack of adequate infrastructure, domestic supply chain and logistics; high cost of finance; inadequate availability of quality power; limited design capabilities and focus on R&D by the industry; and inadequacies in skill development. There is a need for a mechanism to compensate for the manufacturing disabilities vis-à-vis other major manufacturing economies."

And this is just the tip of the iceberg. The government may have been trying to address big companies through such statements, but when it comes to start-ups and MSMEs, these problems amplify! The biggest problem starts with finances, then comes the availability of testing and certification facilities, followed by the availability of components they need, market launch strategies, and various other things.

"The manufacturing disability vis-a-vis China, South Korea, Japan, and some other countries is one of the biggest challenges electronics start-ups face. I think with the new government policies in place a lot of these challenges will be addressed aptly," feels Keskar.

"I think we are all set to create history. We missed the industrial revolutions earlier but as we enter this new technology revolution that started with the invention of electronics, we are on a point where India can bounce back into glory. We have the right talent. We are the youngest nation in the world, and we will remain like that for the next two to three decades. This is our time," feels Sanjay Gupta of NXP Semiconductors.

The first and foremost point that the electronics start-ups and the associated ecosystem should look to capture is the creation of IPs in the country. IP creation, as per Gupta and Keskar, is the heart of anything electronic.

An initiative of MeitY to create a 'fund of funds,' as per a report by Invest India, will work with venture capitalists (VCs) to promote innovation, IP creation, research and development (R&D), and commercialisation. \$101.38 million is committed through EDF to eleven 'Daughter Funds,' which in turn will provide risk capital to companies developing new technologies in areas including electronics, nano-electronics, and information technology. The targeted corpus from EDF outlay is \$846 million.

Additionally, to position India as a global hub for ESDM and push further the vision of the National Policy on Electronics (NPE) 2019, three schemes—namely the Production Linked Incentive Scheme (PLI), Scheme for Promotion of Manufacturing of Electronic Components and Semiconductors (SPECS), and Modified Electronics Manufacturing Clusters Scheme (EMC 2.0)—were notified by the government on April 1, 2020.

"IP is one thing but eventually a system-on-chip (SoC), or IC in simple words, is the second overlap over IP. If you can develop IP, there are tremendous opportunities to develop these ICs or ASICs. These are used to solve application-specific problems. We have talent from India that is developing these ASICs for global giants, but we do not have any Indian company developing that in or outside India," explains Gupta.

The third biggest opportunity for start-ups in the electronics field, as per Gupta, is the ATMP facilities enabling testing and packaging. Companies located near India—including Thailand, China, and Singapore—are super active in establishing such ATMP facilities. However, India seems to be missing this bus as well.

"We are just toddlers into the field of ATMP facilities. I think if we can establish good ATMP facilities, we will be able to add another 20 to 30 percent value in the products," feels Gupta.

These three opportunities, as Gupta explains, are not just opportunities for electronics start-ups, but these are the challenges that, once addressed, will help enable the dream of making India a hub of electronics manufacturing for the world. And if